

AUDITORY TRAINING

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- Auditory training (AT) **is a process of teaching the child with HL to take full advantage of available auditory cues.**
- AT uses everyday activities to foster the development of listening skills.
- Listening is a learned behavior and can be improved with practice!
- The better a child's parents/care providers understand her listening abilities, the more natural and spontaneous activities can become.

AUDITORY TRAINING CONT.

- Utilizes **residual hearing**
- Emphasizing that the primary goal is training the mind to be aware of, attend to, and use **sound**.
- To help integrate **listening** into one's development of communication and social skills
- Key component of AR for CI recipients, those with **prelingual** HL, and those with severe to profound losses

LEARNING TO LISTEN

- A child **must learn to listen** using their amplification devices (hearing aids, cochlear implants).
- Merely providing hearing instruments does not mean the sounds will automatically be **perceived** or **interpreted** by the child.
- By learning to detect and interpret the sounds, the child's capacity for spoken language is **maximized**.

5 EARLY-INTERVENTION PROGRAM

- Ideally begins as soon as the permanent hearing loss is **confirmed**.
- Families may receive a list of all of the available programs in the geographic area and/or be referred to a nation-wide early-intervention system
- Types of Programs
  - **Center-Based Program**: children attend weekly therapy for a set number of hours
  - **Home-Based Program**: early-intervention specialist visits the infant's home and provides instruction to the parents and child



ASSESSMENT OF AUDITORY SKILLS

- Determine if training is **warranted**
- Provide a basis for **post-therapy assessment**
- Identify specific areas of auditory training to concentrate on.

NATURE OF TESTING DEPENDS ON:

- **Age** of child
- His or her **Language skills**
- Type and **degree** of hearing loss

ASSESSING YOUNG CHILDREN

- Both amount and difficulty of testing appropriate for young children are limited by their physical and cognitive development
- **Informal testing and observations** are relied on heavily for this age group
- Initial goal for auditory training of infant may not center on **speech perception**. Rather the focus is on if **hearing** is working appropriately as well as identifying if **auditory skills** have emerged (gross discrimination and localization of a variety of stimuli)
- Parent questionnaires (ex IT-MAIS, PEACH, FAPI) help provide framework for how hearing technology is working and whether programming changes are needed

FAPI

DOCUMENT ON MOODLE

- The Functional Auditory Performance Indicators (FÁPI) assesses the functional auditory skills of children with hearing loss. It can be used by parents, therapists, early interventionists, and teachers. The profile lists auditory skills in an integrated hierarchical order in 7 categories.

FORMAL ASSESSMENTS FOR CHILDREN:

- *WIPI
- ***NU-CHIPS**
- Ling 10 Sound Test
- **Early Speech Perception Test (ESP)**
- Glendonald Auditory Screen Procedure (GASP)
- Developmental Approach to Successful Listening (DASL)

**Additional word recognition assessment information available on MOODLE*

WIPI

- Word Intelligibility by Picture Identification
- Includes only vocabulary appropriate for children with **hearing loss**
- **4** lists, **25** monosyllabic words each
- Child provides picture pointing response in a closed set format
- Suitable for use with children with hearing loss and limited **receptive and expressive** language abilities

NU-CHIPS

- Northwestern University Children's Perception of Speech
- 50 monosyllabic nouns scrambled to form four individual tests
- Response format requires child to point to the one picture from several options that best represent the test item
- **Basic vocabulary; nonverbal response format**

LING 10 SOUNDS

- A behavioral listening check
- The sounds indicate a child's ability to detect all aspects of speech as they encompass the frequency range of all phonemes.
- This check can be used to determine what sounds the student is able to detect, discriminate, and identify

LING 10 CONTINUED

- Should begin sitting next to the child, move to **3** feet then to **6** feet.
- By including the absence of a sound it teaches the child to indicate when they don't hear anything.
- A child with normal hearing can hear all of these sounds at twenty feet.
- Remember to present these in **varying order**.

ESP: EARLY SPEECH PERCEPTION TEST

- Assess pattern and word perception using a **closed-set** format
- Results allow for placement into 4 speech perception categories:
  - No pattern perception
  - **Pattern perception**
  - Some word identification
  - Consistent word identification

GASP – GLENDONALD AUDITORY SCREEN PROCEDURE

- Based on **Erber's** model
- 3 subtests of speech perception:
  - Phoneme detection
  - **Word identification**
  - Sentence comprehension

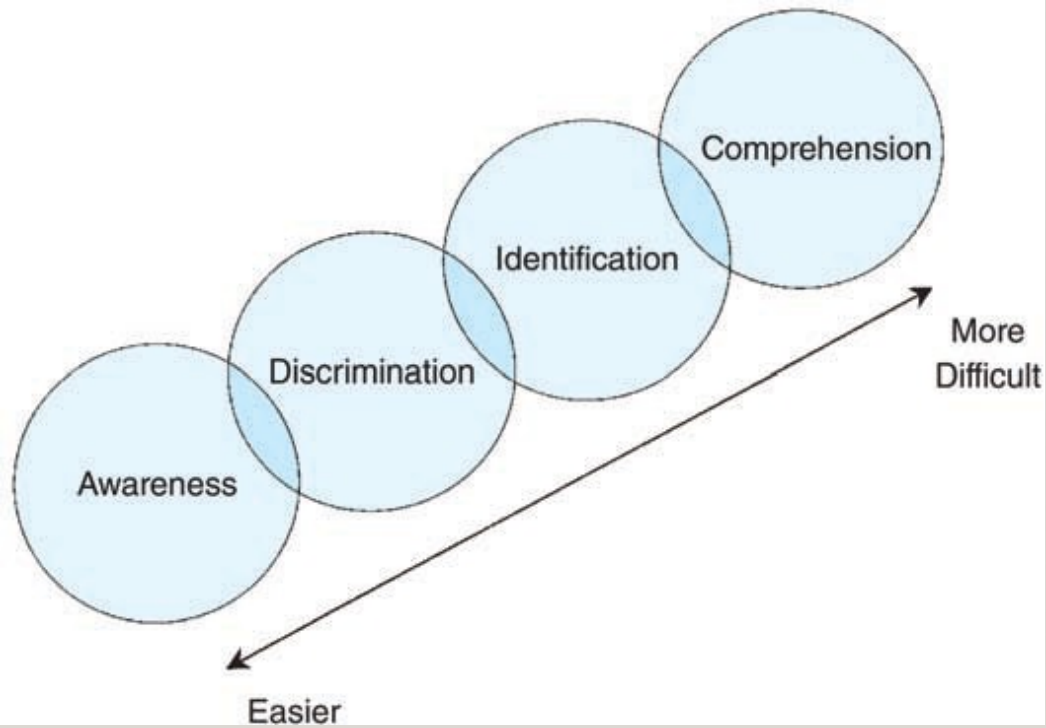
- Results from GASP can aid in planning auditory training because the child's performance on the subtests is predictive of other related auditory tasks

DASL II: DEVELOPMENTAL APPROACH TO SUCCESSFUL LISTENING

Comprehensive test of auditory skills that evaluates:

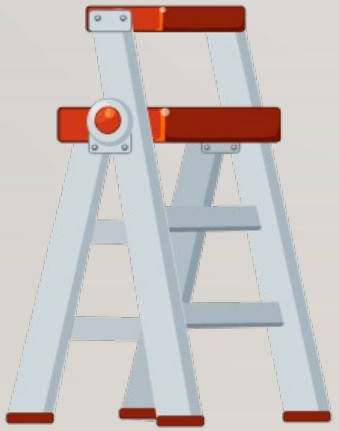
- **Sound awareness**
- Phonetic listening
- Auditory comprehension
- Ages **3** and up
- Some normative information available for children with varying degrees of hearing loss

AUDITORY TRAINING FOR INFANTS AND TODDLERS



BASIC STAGES OF AUDITORY DEVELOPMENT

HIERARCHY OF LISTENING SKILLS/LEVELS OF PERCEPTION MODEL (ERBER)



Comprehension

Identification

Discrimination

Detection

DETECTION OF SOUNDS (SOUND AWARENESS)

- Ability to determine the presence or absence of sound (audition-only).
- "**Sound Conditioning**"

DETECTION CONT.

- To develop sound awareness, a conditioned response is needed.
  - Conditioned response. Child listens and responds in a defined way each time he/she hears a sound.
  - To establish a conditioned response:
    - signal "**listen**"
    - **pause** and establish eye contact
    - **cover mouth**, be careful not to muffle the sound
    - pause (vary length to avoid over anticipation)
    - provide the sound
    - expect child to respond (may have to guide the child at first)

EXAMPLES OF DETECTION TASKS

- responds to loud, low frequency environmental sound (e.g., drum)
- responds to presence of a **speech syllable** (e.g., /ba/)
- responds to variety of environmental sounds (e.g., whistle, horn, knock on the table)
- indicates when an ongoing **environmental sound stops** (e.g., music)
- indicates when a continuous speech syllable stops (e.g., sustained /ba/).

23 DETECTION/SOUND AWARENESS ACTIVITIES



- Play *peek-a-boo*
- Play musical chairs
- March to the beat of a drum
- Push the toy care whenever the clinician says “Vrrrrm”

EXAMPLE: DETECTION OBJECTIVE

- Detects strong and/or low-frequency voice 1-3 feet from CI microphone.
- Attends to 3 loud toy or environmental sounds from 4-6 feet from FM microphone.
- Detects 3 moderate intensity environmental sounds (e.g., baby swing music, microwave beep to indicate bottle is ready, running water for a bath).

Taken from: *Parent-Infant Communication*

DISCRIMINATION

- Discrimination of sounds refers to the ability to perceive similarities and differences between two or more sounds (including speech stimuli).
- Can include **suprasegmental and segmental contrasts (duration / intensity / pitch)**

Taken from: Sandie Bass-Ringdahl, University of

EXAMPLES OF DISCRIMINATION TASKS

- repeats the correct number of syllables heard
- discriminates between **loud and soft sounds and durations.**
- discriminates between repeated syllables using a fast rate versus a slow rate
- discriminates between **high- and low-pitched sounds (whisper / quiet / loud speech)**
- discriminates between sound continuously varying from high to low, and low to high pitch
- repeats vowel sounds

DISCRIMINATION ACTIVITIES

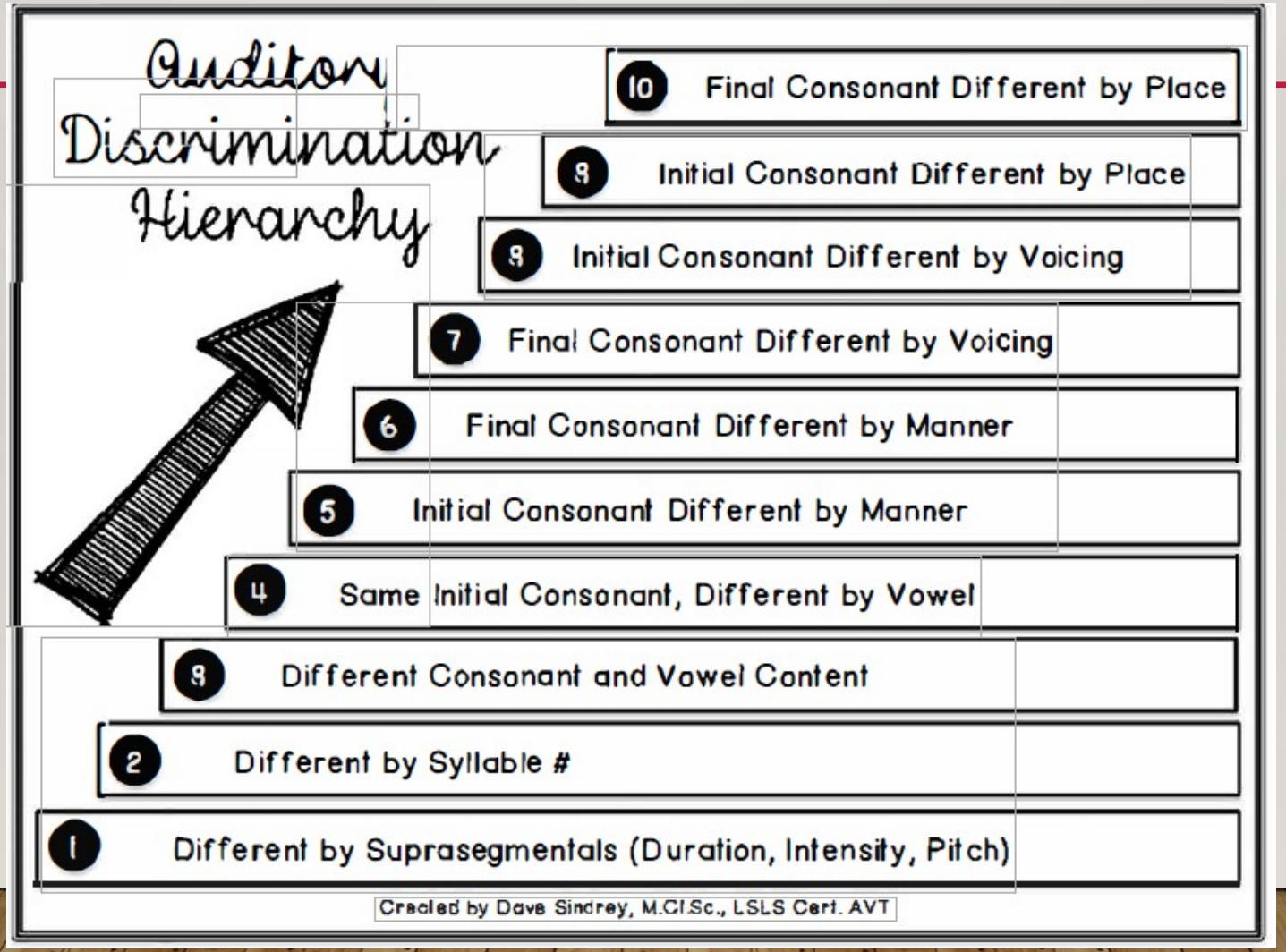
- Play a game with toy animals (“The cow says ‘moo’; the sheep says
- ‘baaa’ ”)
- Respond to commands (“Clap your hands”; “Jump!”)
- Play a *Same* or *Different* game (“car car”; “car star”) using pairs of picture cards where the child has to point to the correct illustration
- Repeat what you hear (“Mamma,” “Papa”)

EXAMPLE: DISCRIMINATION OBJECTIVE

- Discriminates voice from primary caregiver from other voices (e.g., smiles when mom speaks).
- Discriminates angry voice from happy voice (e.g., baby cries when angry voice speaks and smiles with happy voice).
- Discriminates two vocalizations (e.g., baby imitates mom's "mm", but does not vocalize at mom's "babababa").

Taken from: *Parent-Infant Communication*

AUDITORY DISCRIMINATION HIERARCHY



30 IDENTIFICATION

- Ability to **repeat, point to, or write the speech stimulus heard**

Taken from: Sandie Bass-Ringdahl, University of

EXAMPLES OF IDENTIFICATION TASKS

- words varying in number of syllables
  - hat vs. ice cream cone
- one syllable words varying in **vowels and consonants**.
  - bat vs. fish
- stereotypic messages (familiar expressions and directions)
- words with identical consonants and different vowels
  - book vs. bear
- words with identical vowels and consonants differing **in place, manner, and/or voicing**
  - dog vs. fog

EXAMPLE: IDENTIFICATION OBJECTIVE

- Recognizes an adult-produced animal sound and associates sound with toy animal.
- Identifies 3 loud instrument sounds: drum, guitar, and bells.
- Recognizes 10 words that label common objects (e.g. bottle, diaper, ball).

Taken from: *Parent-Infant Communication*

IDENTIFICATION



- Play the game *Candy Land* and listen for the names of the colors
- Play with sets of postcards or stickers ("Show me the cat")
- Play *Go Fish* with cards ("Give me your sevens"; "Give me your twos")

COMPREHENSION

- Comprehension of sounds refers to the ability to understand the meaning of speech by **answering questions, following directions**, or participating in a conversation.
  - The child's response must be qualitatively different than the stimuli presented!
- This level requires that the person have a certain degree of **syntactic and semantic** language development

Taken from: Sandie Bass-Ringdahl, University of

EXAMPLES OF COMPREHENSION TASKS

- Auditory sequencing
  - familiar expressions
  - follow single directions
  - follow classroom directions
  - follow multi-element directions
- Structured listening set
  - recall details of a story
  - understand the main idea of a lesson or complex story
- Conversation
  - answer questions requiring comprehension of the main idea
  - paraphrase remarks of another
  - offer spontaneous relevant remarks

Taken from: Sandie Bass-Ringdahl, University of Iowa

EXAMPLE: COMPREHENSION OBJECTIVE

- Demonstrates comprehension of 2 phrases through audition alone (e.g., Do you want up?).
- Comprehends increasingly complex requests (e.g., bring me your socks and shoes).
- Comprehends and responds appropriately to sentences in which possessive nouns or pronouns are contrasted (e.g., Where are you eyes? vs. Where are mommy's eyes?).
- Comprehends the meaning of 3, loud environmental sounds (e.g., smoke alarm, doorbell, car horn).

Taken from: *Parent-Infant Communication*

ILLUSTRATIVE FORMAL AUDITORY TRAINING LESSON PLAN

- ***Title:*** Snake and Ice Cream Game
- ***Goal:*** Discrimination
- ***Objective:*** The student will discriminate a nasal consonant versus non-nasal unvoiced consonant that differs in place of production.
- ***Materials:***
  - a picture of a snake to represent the /s/ sound
  - a picture of a boy about to eat an ice cream sundae to represent the /m/ sound
  - a stack of 26 pennies for reinforcements

- ***Procedures:***
- Introduce the picture of the snake by pointing to it and saying "sssssssss . . ." Ask the child to imitate your production. Similarly, introduce the picture of the sundae, and say "mmmmmm . . ." Ask the child to imitate your production.
- Say each sound with your face visible. After each utterance, ask the child to point to the corresponding picture.
- Place 13 pennies on each picture. Cover your mouth. Randomly say one sound after another. After each production, the child may pick up a penny from one of the two pictures. If the child removes a penny from the correct picture, he or she can keep it. If incorrect, the penny must be placed back on the picture.
- Continue until all pennies are spent.

AUDITORY TRAINING APPROACHES IN 4 CATEGORIES:

- Analytic
- **Synthetic**
- Pragmatic
- **Eclectic**

ANALYTIC AUDITORY APPROACH

- Break speech into smaller components
- **(same - different) Bottom-up approach**
- Vowels vs consonants

ANALYTIC THERAPY OBJECTIVES

- Will discriminate vowels that differ in first formant information Will discriminate vowels that differ in second formant information
- Will discriminate words that have vowels with similar first and second formant information
- Will identify words with different vowels, using a four-item response set
- Will identify words with different vowels, from an open set of vocabulary

SYNTHETIC

- Emphasizes use of clues derived from syntax and context
- Use of meaningful stimuli
- **(Sentence perception) Top-down approach**

SYNTHETIC THERAPY OBJECTIVES

- Will discriminate between a declarative and interrogative sentence
- Will carry on a conversation
- Can follow the instructions of simple commands

PRAGMATIC

- Training listener to control communication variables
- Level of speech
- **Signal-to-noise ratio (SNR)**
- Context or complexity of message
- Table 4.8 QUEST?AR

ECLECTIC:

- Combines most or all of the strategies

GOALS OF AUDITORY TRAINING

- There are many factors that can be manipulated to make a task **easier or harder**
- Regardless, the goal is always to make the listening task challenging without being frustrating.
- The most useful learning events are those that take place in a **meaningful context for that child specifically.**

FOUR DESIGN PRINCIPLES

1. Auditory Skill
   - Awareness, detection
   - Discrimination
   - Identification
   - Comprehension
2. Stimuli
   - Phonetic level
   - Sentence level
3. Activity Type
   - Formal
   - Informal
4. Difficulty Level
   - Response type
     - Closed, limited, open
   - Stimulus unit
     - Words, phrases, sentences
   - Stimulus similarity
   - Contextual support
   - Task structure
     - Highly structured, spontaneous
   - Listening conditions

3. ACTIVITY TYPE

- **Informal**
  - Occurs as part of the daily routine
  - Often incorporated into other activities
- **Formal**
  - Highly structured
  - Usually one-on-one training

DIFFICULTY LEVEL

- Stimulus set----------
- Stimulus unit----------
- Stimulus similarity------
- Contextual support----
- Task structure----------
- Listening conditions (SNR)



FORMAL AUDITORY TRAINING PROGRAMS

- There are a number of formal, commercially available auditory training programs available for use.
  - *SPICE (Speech Perception Instructional Curriculum Evaluation)*
  - *Learn To Talk Around The Clock*
  - *Cottage Acquisition Scales For Listening, Language, and Speech*
  - *Developmental Approach to Successful Listening (DASL)*
  - *SKI-HI*

DASL II

- **Developmental Approach to Successful Listening II**
- **Consists of hierarchy of listening skills worked on in brief, individualized sessions**
- **Main use is with preschool and school age using hearing aids/cochlear implants**
- **Emphasizes 3 areas of auditory skill development: Sound Awareness, Phonetic Listening and Auditory comprehension**

SPICE

- Speech Perception Instructional Curriculum and Evaluation
- Used for evaluating and developing auditory skills in children with severe to profound hearing loss
- Contains goals and objectives associated with 4 levels of speech perception: Detection, Suprasegmental, Vowel and Consonant Perception, and Connected Speech

SKI HI

- **Sensory [Kids] Impaired Home Intervention**
- **Comprehensive identification and home intervention curriculum for infants with hearing loss and their families**
- **Developmentally based auditory stimulation training program**
- **Used in conjunction with language-speech stimulation and consists of four phases and eleven general skills**
- **Table 4.10**

COTTAGE ACQUISITION SCALES FOR LISTENING, LANGUAGE, AND SPEECH

- Age range: Birth-8 years
- Integrated approach to language intervention by providing a single set of instruments for following the development of language, listening, cognition and speech in children with hearing loss.
- Skills addressed include: Sound Awareness, Phonetic Listening, Suprasegmental Speech Duration, Vowels and Consonants, Vocalization & Phonetic Level Speech, Phonological Level Speech

AUDITORY TRAINING PROGRAMS

- Choose the AT program that best meets the child's and the family's needs.
- All of the programs are structured using the same basic principles—**the hierarchy of listening skills- Erber's levels of perception**

GOALS

RESOURCES

- John Tracey Clinic
- Central Institute for the Deaf (creators of the SPICE curriculum)
- BRIDGE Program (Med EL)
- The Listening Room (Advanced Bionics)
- Sound Foundations (Cochlear Americas)
- AG Bell
- The SKI-HI Institute